HEISENBERG UNCERTAINTY PRINCIPLE ASSIGNMENTS

1. (a) If the position of an electron in an atom is measured to an accuracy of 0.0100 nm, what is the electron's uncertainty in velocity?
   (b) If the electron has this velocity, what is its kinetic energy in eV?

2. The position of a chloride ion on a material can be determined to a maximum error of 1μm. If the mass of the chloride ion is 5.86×10^{-26}kg, what will be the error in its velocity measurement?

3. The lifetime of an excited state of an atom is 3×10^{-3}s. What is the minimum uncertainty in its energy in eV?

4. Determine the minimum uncertainties in the positions of the electron if their speeds are known with a precision of 3.0×10^{-3}m/s